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Class : SYIT	Batch : -
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PDS Practical - IX

```
import pandas as pd
```

```
# 1. Import a CSV dataset using Pandas
df = pd.read_csv("data.csv")
print("Original Dataset:")
print(df)
```

Original Dataset:

	Name	Age	Marks	City
0	Rahul	20.0	85.0	Mumbai
1	Priya	NaN	90.0	Delhi
2	Amit	21.0	NaN	Pune
3	Sneha	19.0	88.0	Mumbai
4	Rahul	20.0	85.0	Mumbai
5	Arjun	NaN	75.0	Chennai

```
# 2. Find missing values in the dataset
print("\nMissing values in each column:")
print(df.isnull().sum())
```

Missing values in each column:

```
Name      0
Age        2
Marks      1
City       0
dtype: int64
```

```
# 3. Replace missing numerical values with the mean
num_cols = df.select_dtypes(include="number").columns
df[num_cols] = df[num_cols].fillna(df[num_cols].mean())

print("\nDataset after replacing missing values with the mean:")
print(df)
```

Dataset after replacing missing values with the mean:

	Name	Age	Marks	City
0	Rahul	20.0	85.0	Mumbai
1	Priya	20.0	90.0	Delhi
2	Amit	21.0	84.6	Pune

```
3 Sneha 19.0 88.0 Mumbai
4 Rahul 20.0 85.0 Mumbai
5 Arjun 20.0 75.0 Chennai
```

```
# 4. Remove duplicate records
print("\nNumber of duplicate records:", df.duplicated().sum())
df = df.drop_duplicates().reset_index(drop=True)
```

Number of duplicate records: 1

```
# 5. Display the cleaned dataset
print("\nCleaned Dataset:")
print(df)
```

Cleaned Dataset:

	Name	Age	Marks	City
0	Rahul	20.0	85.0	Mumbai
1	Priya	20.0	90.0	Delhi
2	Amit	21.0	84.6	Pune
3	Sneha	19.0	88.0	Mumbai
4	Arjun	20.0	75.0	Chennai